Course Notes 5030

Decisions, Games, and Markets

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Chapter 1

Introduction

1.1 Purpose of the Course

The goal is to give you the theoretical tools to structure complex problems so that you can use and analyze data to answer meaningful questions.

Particularly, I want to give you the theoretical tools to (i) be critical of empirical methods and understand their misapplication, (ii) come up with well-reasoned answers to difficult questions when data is scarce, and (iii) utilize theoretical understanding to structure problems that are otherwise problematic.

I hope this course will teach you to avoid the blind application of econometric techniques without understanding the appropriateness of the tool.

1.2 What Is Economics?

Before moving toward more specific material, we are going to present and reconsider some foundational ideas in economics.

Let us think about two related definitions of economics:

Definition 1.1. The science of dealing with scarcity problems related to production, distribution, and consumption of goods and services.

Definition 1.2. The science of understanding positive and normative impacts of human decisions regarding problems of scarcity.

In modern applications, economics is an incredibly broad field of social science. The field covers problems that are traditionally thought of as "economic," but also includes overlapping content with all other fields of social science. The tools of economic analysis are also applied to, and overlap with, many "hard" scientific fields, such as physics, biology, epidemiology (medicine in general), neuroscience, computer science (artificial intelligence), and ethology.

1.3 What Should We Care About with Regard to an Economic System?

Economic activity can be broken down into two basic components: production and allocation. Production is simply the process of transforming some inputs into a good or service. It is essentially the process of taking what is already present and creating something new.

Allocation is the process of who gets what in terms of goods and services. Any economic activity falls within an economic system.

Definition 1.3. An economic system, or economic order, is a system of production, resource allocation, and distribution of goods and services within a group of humans.

This includes the combination of the various institutions, agencies, entities, decision-making processes, and patterns of consumption that comprise the economic structure of a given community.

As we begin our journey toward a deeper understanding of economic ideas, we first should ask ourselves, "What should we care about knowing about an economic system?" In other words, what is it that we would like to be able to describe, understand, and predict?

The traditional answer to this question can be broken into two categories of analysis:

Definition 1.4 (Positive Economic Analysis)**.** Positive economic analysis focuses on describing and predicting outcomes in any economic setting. Positive analysis tries to answer the questions: What is? and What will be?

This type of analysis is in accord with the analysis done in the physical sciences. For example, in physics they focus on describing and predicting outcomes in the physical world. This type of analysis in economics is like the physical sciences and uses many related mathematical tools.

Definition 1.5 (Normative Economic Analysis)**.** Normative economic analysis focuses on evaluating outcomes as better or worse. This analysis looks to answer the question: What should be?

Normative economic analysis is a type of philosophy, making the applications of tools from the physical sciences much less appropriate. Inherently, normative analysis is subjective, but this does not mean we cannot work to understand it further. In fact, there is much to gain by just defining basic concepts and language for normative analysis.

With regard to economic outcomes, the two classical normative concepts, or "measures," are equity and efficiency.

Definition 1.6 (Equity)**.** Equity is a broad term meaning fairness; this can be defined in many ways in an economic setting, but clearly it is a concept that we should at least consider in evaluating an economic system.

Understanding based on who a person is (human capital, physical capital), what they do in an economic system, and what they will receive are fundamental methods to evaluate the system.

Efficiency is another basic concept we use in economics to evaluate outcomes. Loosely, efficiency can be broken into two categories: productive efficiency and allocative efficiency.

Definition 1.7 (Productive Efficiency)**.** Productive efficiency is when a system cannot possibly produce one more good or service without sacrificing some production of another good or service. In a narrower sense, this implies that production is done at minimum cost.

Definition 1.8 (Allocative Efficiency)**.** Allocative efficiency is, given a fixed set of produced goods and services, the condition in which the person who gets each good or service values it more than anyone else would be willing to pay for that good or service.

These two concepts come together in the most important concept of efficiency used in economics due to Vilfredo Pareto:

Definition 1.9. An outcome is Pareto efficient as long as there is no way to change the production or allocation to make someone better off without hurting anyone else.

Together, the concepts of equity and efficiency provide us with two useful dimensions to construct an evaluation of what is "better." To be clear, this is not saying these are the only relevant dimensions, but instead that these two are the classic means of evaluation and are useful. We will discuss and explore other dimensions for normative evaluation as they apply to our various subjects. (For example, the dimension of "freedom" will be discussed in our chapter on game theory.)

1.4 Classifications of an Economic System

We will explore thinking about an economic system in a number of ways. First let us describe a system based on its level of central control. Two extreme ends of the spectrum are:

Definition 1.10 (Pure Command and Control System)**.** In a pure command and control system all production and consumption is determined by a central planner. There are no decentralized interactions—no markets.

Definition 1.11 (Pure Decentralized System)**.** In a pure decentralized system all economic activity (production and consumption) is determined by decentralized interactions (only markets).

It is worth noting that both systems have obvious problems. The obvious problems with command and control are clear in any large economic system. The central planners, even if well intentioned, will not have the necessary information to make decisions that can lead to

any concept of efficiency. The obvious problem with pure decentralization is that there are no rules, i.e., nothing helping the economy move toward efficiency or equity. This pure form of laissez-faire is an anarchic system without a justice system (no centralized enforcement of property rights).

We will learn that extremes are highly problematic in complex settings. Furthermore, these ideas do not correspond well with popular notions of economic systems (that are usually poorly defined). These popular notions I call the system "isms." These are capitalism, socialism, and fascism. What are these ideas in terms of economic systems? And how do they fit in the decentralized vs. centralized framework?

Most economic systems are far too complex to fit into one of these categories alone. All real-world economic systems have a mixed allocation of property rights and industrial policies that favor various actors.

1.5 Brief Descriptive Outline of the Course

I have provided all of these basic definitions to give us a foundation to think about economic theory: what problems should we address and understand in economic models? We now move to building economic theory from the ground up. Eventually, we will come back to the far more detailed analysis of economic policies and systems using the tools we develop.

Chapter 2

Science of the Artificial

The title of this chapter borrows the title of Herbert Simon's seminal book, as the "artificial" problem of representing something in a person's head, "preferences," in a language that allows for reasoning about decisions is the subject of the chapter. Before delving into preferences and decisions, we will provide some base understanding of measurement using binary orders. This discussion is valuable in its own right, as it elucidates very simple, but often overlooked, problems with econometric analysis.

2.1 Foundations of Measurement

We present a series of examples that illustrate basic measurement issues.

Example 2.1 (Height)**.** Take a structure in the physical world and we will define its height as the greatest perpendicular distance from that structure to the ground it sits upon. Thus, this is a measure of any physical structure. Consider different units of measurement for this distance: inches and centimeters. Notice that these two different measures are simply a

linear transformation of one another. That is, 1 inch equals 2.54 cm.

Thus, the cardinalities (values) of the measures are related by a constant. In this case these two measures are invariant to standard statistical operations. For example, we can take the mean of a set of heights measured in these two ways and they will also have the same cardinal relationship. To show this, take the example of three objects measured in inches and centimeters:

$$I = \{10, 20, 60\}, \qquad E[I] = 30$$

$$C = \{25.4, 50.8, 152.4\}, \qquad E[C] = 76.2$$

$$2.54E[I] = E[C].$$

To take this further, if we use one measurement or the other in an OLS regression, the estimates will be cardinally related by the constant 2.54 and statistical significance is unaffected.

Example 2.2 (Strength of an Earthquake)**.** One measure of earthquake strength is the maximal amplitude A (simplified). The second measure of an earthquake's strength is the Richter scale, which is $\log(A)$. These two measures provide the same ordinal relation; that is (within the set of measurable objects) $A > A'$ if and only if $\log A > \log A'$. This is true for any measurement system based on a strictly increasing function (monotonic). But note, we lose the linear cardinality relationship of the measures for this example. This turns out to be quite important for statistical analysis in that replacing one measure with the other does not result in easily transformable statistics.

For example,

$$A = \{1, 10, 13\}, \qquad E[A] = 8$$

$$\log A = \{0, 1, 1.12\}, \qquad E[\log A] = 0.707$$

$$\log E[A] = 0.903 \neq E[\log A] = 0.707.$$

This means an OLS regression using A and $\log A$ will have non-transformable statistics. You could in fact have positive significant results in one case and non-significant in the other.

Example 2.3 (Abandoned Structures)**.** A researcher cares about the number of abandoned buildings in a city, but uses vacancy rate as a proxy measure. More precisely, this researcher wants to know the relationship between building abandonment and crime to assess a city's policies of destroying abandoned buildings. But the only data available for a related measure is vacancy rate.

Vacancy rate is the percentage of unoccupied buildings in an area, while abandonment rate is the percentage of abandoned buildings in an area. Clearly there are many buildings that are not occupied but not abandoned, and there could be occupied buildings that are abandoned. So across cities these two measures might not even preserve the same order, let alone have a linear cardinality relationship. This begs the question: how can we interpret statistics done based on vacancy rate for the true variable of interest, abandonment rate? The answer is not immediately clear and must come out of a deeper understanding of the relationship between these measures. Many empirical research questions run into such a problem. Without a clear understanding of the relationship between the measurements, interpretation of the statistical results will be problematic.

The lesson is that a numerical representation contains only the information preserved by its permissible transformations.

Definition 2.4. A numerical representation is **ordinal** if only the ordering of objects is meaningful. If u is an ordinal representation, then $f \circ u$ represents the same ordering for every strictly increasing function f.

Example 2.5. Suppose a student ranks tea above coffee and coffee above water. Assigning the values 3, 2, and 1, respectively, captures that ranking. Assigning 30, 20, and 10 captures exactly the same information: the gaps between the numbers do not have an independent meaning.

Definition 2.6. A representation has **positive-affine cardinal structure** if the relevant comparisons are preserved only by transformations

$$v = au + b, \qquad a > 0.$$

Example 2.7. If two outcomes have utility values 4 and 1, then the transformation $v = 2u+3$ gives them values 11 and 5. The new numbers preserve the comparisons relevant for expected utility, including comparisons between lotteries over the two outcomes.

This distinction will matter repeatedly. Ordinary deterministic utility maximization is ordinal. Expected utility requires more structure because probabilities multiply numerical utility differences.

2.2 Representation of Preferences

The presentation of the measurement problem in the previous subsection has two purposes: (i) to highlight how important measurement is for interpretation of statistical analysis, and (ii) to set the stage for the abstract example of representing a person's preferences in mathematical language, which is an abstract measurement problem.

We will take a person's preferences over a set of objects (choices, outcomes, etc.) as a binary order on objects in that set. We will see that the easily representable properties of this preference operator are going to be only ordinal. The following example should clarify this point.

Example 2.8. Consider preferences over three lunch items: hot dog, cheeseburger, and burrito.

Tony has the following preferences: hot dog is at least as good as a cheeseburger, hot dog is at least as good as burrito, and burrito is at least as good as cheeseburger.

Cindy has the following preferences: cheeseburger is better than hot dog, hot dog is better than burrito, and burrito is better than cheeseburger.

Bill has the following preferences: cheeseburger is better than hot dog and he cannot compare anything to burrito because he does not know what a burrito is.

Notice if we ask these people to give numeric values to each food item, the only information we can take from these values is the order. Without additional information, the cardinality (numeric values) provides no other insights beyond the usable insights on the person's preferences.

We will refer back to this example throughout the more abstract presentation of key concepts below.

2.2.1 Abstract Conceptualization

Formally, a preference relation is a partially ordered space. That is, the pair of a partial order $\succeq$ and the set S. We write this pair simply as $(\succeq, S)$. Both parts are important because the order is only well-defined on the space S. The partial order (or preference relation) $\succeq$ is defined on elements of the set S; it provides a binary relation between two elements. That is, the mathematical notation $x \succeq y$ means that x is "weakly preferred" to y or "at least as good." It is standard to think of this as the preference of a single decision maker, although other interpretations such as the preference of a group are also useful.

Using $\succeq$ as our basis, we define other binary relations on S. First define "not weakly preferred" as $x \not\succeq y$, which means that the relation $x \succeq y$ does not hold. The binary operator "better than" is defined as $x \succ y$ if $x \succeq y$ and $y \not\succeq x$. The relation "equivalent to" is defined as $x \sim y$ if $x \succeq y$ and $y \succeq x$.

Now let us use the preference relation to define the optimal choice set of an individual. These are the set of choices that a person weakly prefers to every other choice in the set. Formally, this optimal choice set $C(\succeq, S)$ is defined as follows:

$$C(\succeq, S) = \{x \in S : x \succeq y \text{ for all } y \in S\}.$$

Example 2.9. If a diner ranks the three meals on a menu and regards pasta as at least as

good as both soup and salad, then pasta belongs to the optimal choice set. If soup and pasta are tied for best, both belong to that set.

Note that the set of optimal choices can be empty. When it is not empty, this is the set of choices that follow an individual's preferences.

Core properties of preferences. The two most basic properties of preferences are formalized below. The first property is that a person can compare everything in the set S with the preferences $\succeq$. This is called completeness. In Example 18, both Tony and Cindy have complete preferences over the three lunch choices while Bill does not.

Definition 2.10. The preferences $(\succeq, S)$ are complete if for all $x, y \in S$, either $x \succeq y$ or $y \succeq x$ (or both).

Example 2.11. If a person is choosing among tea, coffee, and water, completeness means that the person can compare every pair. For example, the person can say whether tea is at least as good as coffee, even if the answer is that the two are equally good.

The second fundamental property of preferences is consistency across choices. This is called transitivity. On one hand, this property is reasonable if we do not consider preferences to be fixed over time (or equivalently the choice set at time t is all different objects than the choice set at time t'). On the other hand, assuming this type of consistent behavior over time is a very strong assumption about human behavior. In Example 18, only Tony has transitive preferences. It is worth noting that transitivity does not require completeness.

Definition 2.12. The preferences $(\succeq, S)$ are transitive if for all $x, y, z \in S$, if $x \succeq y$ and $y \succeq z$, then $x \succeq z$.

Example 2.13. If a person ranks tea at least as good as coffee and coffee at least as good as water, transitivity requires the person to rank tea at least as good as water. A cycle in which tea is preferred to coffee, coffee to water, and water to tea would violate transitivity.

2.2.2 Finite Set of Choices

Let us consider the case where the set S is finite. That is, there are a finite number of objects in S.

For this case, we first show that a complete and transitive preference relation has a set of optimal choices. Second, we show that under these same assumptions the preferences can be represented by a real-valued "utility function" and that maximizing this utility function is an alternative path to the optimal choice set.

Proposition 2.14. *Consider the case that S is finite and that $(\succeq, S)$ is complete and transitive. Then $C(\succeq, S) \neq \varnothing$.*

Example 2.15. Consider a menu with three meals. If a diner can compare every pair of meals and ranks them consistently, at least one meal is at least as good as every other meal. That meal belongs to the optimal choice set.

Back to Example 18, we can see that for Tony the optimal choice set contains hot dog (we would need more information to know if it contains other options). Bill has an empty optimal choice set because nothing is comparable to burrito, a violation of complete preference. Cindy has an empty optimal choice set because there is another option strictly preferred to every option. This is a violation of transitivity. This helps illustrate the importance of both assumptions for the existence of an optimal choice.

Proof. Suppose to the contrary that S is finite, $(\succeq, S)$ is complete and transitive, and $C(\succeq, S) = \varnothing$. We will show a contradiction. Let us write $S = \{x_1, x_2, \ldots, x_K\}$ for an arbitrary finite set with K elements. Since $C(\succeq, S) = \varnothing$, and $(\succeq, S)$ is complete, for each $x \in S$ there exists $y \in S$ such that $y \succ x$. This is because if $x \not\succeq y$, then completeness implies that $y \succeq x$, and this is the definition of $y \succ x$. Let us consider the element x_1 and denote x_2 as an element such that $x_2 \succ x_1$. Then for x_2, denote x_3 as an element such that $x_3 \succ x_2$. Note by transitivity $x_3 \succ x_1$, which makes it a distinct element. We iterate this argument and for the arbitrary element x_k there is an x_{k+1} with $x_{k+1} \succ x_l$ for all $l \in \{1, 2, \ldots, k\}$ by transitivity. Then for element x_{K-1} there is an x_K with $x_K \succ x_l$ for all $l \in \{1, 2, \ldots, K-1\}$ by transitivity. Then for the element we call x_K, $x_K \succeq x$ for all $x \in S$, which means $x_K \in C(\succeq, S)$, a contradiction to this set being empty. □

Now we present the idea of representing the preferences $(\succeq, S)$ with a real-valued utility function $u : S \to \mathbb{R}$. The following definition formalizes the idea.

Definition 2.16. The function $u : S \to \mathbb{R}$ represents the preferences $(\succeq, S)$ if the following holds: for all $x, y \in S$, $x \succeq y$ if and only if $u(x) \geq u(y)$.

Example 2.17. Suppose a consumer ranks apples above bananas and bananas above oranges. The utility assignment $u(\text{apples}) = 3$, $u(\text{bananas}) = 2$, and $u(\text{oranges}) = 1$ represents those preferences because the numerical order matches the consumer's ranking.

It turns out that for the finite case a preference relation that is complete and transitive always has a utility function that represents the preferences.

Proposition 2.18. *Consider the case that S is finite and that $(\succeq, S)$ is complete and tran-*

sitive. Then there exists a $u : S \to \mathbb{R}$ *that represents the preferences* $(\succeq, S)$.

Example 2.19. If a student regards two textbooks as equally good and prefers both to a third textbook, one representation is to assign the first two textbooks utility 2 and the third utility 1. Thus tied choices receive the same number, while better choices receive higher numbers.

Proof. Based on the argument in the proof of Proposition 21, we know there must be a non-empty set $C(\succeq, S)$. Note for all $x, y \in C(\succeq, S)$, $x \sim y$ (since it must be that $x \succeq y$ and $y \succeq x$), and $x \in C(\succeq, S)$ and $z \in S \setminus C(\succeq, S)$ imply $x \succ z$. For convenience, re-label this subset $C(\succeq, S)$ of S as S_1. Now consider the set $C(\succeq, S \setminus S_1)$, which is non-empty by Proposition 21 if $S \setminus S_1$ is nonempty. We can iterate our previous argument and set that for all $x, y \in C(\succeq, S \setminus S_1)$ and $z \in (S \setminus S_1) \setminus C(\succeq, S \setminus S_1)$, $x \sim y$ and $x \succ z$. Denote the set $C(\succeq, S \setminus S_1) = S_2$. We iterate until there are no elements remaining and get a set of subsets $\{S_1, S_2, \ldots, S_L\}$ for some arbitrary finite L less than or equal to the number of elements in the set and such that $\bigcap_{l=1}^{L} S_l = \varnothing$, $\bigcup_{l=1}^{L} S_l = S$, for all $x, y \in S_l$, $z \in S_j$, $x \sim y$, and $x \succ z$ for $l > j$. We use this structure to construct a utility function as follows: for all l, $u(x) = L - l$ if $x \in S_l$, and clearly $u(x) \geq u(y)$ if and only if $x \succeq y$. □

The existence of a utility representation is extremely convenient because it allows us to frame the optimal choice problem as a classic optimization problem. That is, the set of maximizers of the function u on S is the optimal choice set. We prove this in the following proposition.

Proposition 2.20. *Consider that* $u : S \to \mathbb{R}$ *represents the preferences* $(\succeq, S)$. *Then*

$C(\succeq, S) = \arg\max\{u(x) : x \in S\}$.

Example 2.21. With the textbook utilities 2, 2, and 1, the utility-maximizing choices are the two textbooks assigned utility 2. They are also exactly the choices the student weakly prefers to every other textbook.

The proof of this proposition follows immediately from the definitions of $C(\succeq, S) = \{x \in S : x \succeq y \text{ for all } y \in S\}$ and $\arg\max\{u(x) : x \in S\} = \{x \in S : u(x) \geq u(y) \text{ for all } y \in S\}$, and the fact that $x \succeq y$ if and only if $u(x) \geq u(y)$.

Since preferences are only ordinal binary measures, there are many utility representations (infinite) for a set of preferences. In fact, for any utility representation u, any monotonic transformation of u, $v = f(u)$ where $f : \mathbb{R} \to \mathbb{R}$ is a strictly increasing function, also represents the same preferences. This means the cardinality or actual numerical value of the utility function is meaningless except for the order it creates across choices.

2.2.3 Preferences over Commodities

In this subsection we will consider the spaces of choices to be a commodity space $X \subseteq \mathbb{R}^L$, for $L \in \mathbb{N}$. This is a set in which each element is a vector $x \in X$ describing the quantity of all L goods and services. Each $x = (x_1, \ldots, x_L)$ with x_l the quantity of good l. We will replace the general set S in the preference definition with X, which is a subspace of $\mathbb{R}^L$. Thus a preference relation is written $(\succeq, X)$. Preferences over commodity spaces have some additional complexity because X is not a finite space. For this reason, the assumption that $(\succeq, X)$ is complete and transitive is not enough to guarantee the existence of an optimal choice set and a utility representation.

An additional technical condition must be introduced to guarantee that these properties hold. The condition is preference continuity, formally defined as follows.

Definition 2.22. $(\succeq, X)$ is continuous if for any sequence of pairs $\{x^n, y^n\} \subseteq X$ with $x^n \succeq y^n$, $\lim x^n = x$, and $\lim y^n = y$, we have $x \succeq y$.

Example 2.23. Suppose a consumer evaluates bundles by $u(x) = x_1 + x_2$, the total amount of two goods. If a sequence of bundles is always at least as good as another sequence, then their limiting bundles retain that ranking; an arbitrarily small change in quantities cannot create a sudden reversal.

Continuity stops preferences from jumping at limit points and corresponds with the existence of a continuous utility function representing preferences. The following example of lexicographic preferences violates continuity.

Example 2.24. Consider $X = \mathbb{R}^2$ and preferences $\succeq$ defined as $x \succeq y$ if $x_1 > y_1$ or $x \succeq y$ if $x_1 = y_1$ and $x_2 \geq y_2$. Now take the sequences $x^n = (1/n, 0)$ and $y^n = (0, 1)$. Note that for all n, $x^n \succeq y^n$, but $x \not\succeq y$.

Adding continuity is sufficient to extend the utility representation theorem to preferences on the commodity space.

Proposition 2.25. *Let $X \subseteq \mathbb{R}^L$ be compact. Suppose $(\succeq, X)$ is complete, transitive, and continuous, and that a continuous utility function $u : X \to \mathbb{R}$ represents $(\succeq, X)$.*

Example 2.26. Let $X = [0, 1]^2$ and suppose a consumer ranks bundles by $u(x) = x_1 + x_2$. This continuous utility function gives nearby bundles nearby utility values, and its numerical ranking agrees with the consumer's preferences.

The proof of this result is technical and requires the following lemma.

Lemma 2.27 (Ordered-space representation)**.** *Let Y be a nonempty connected separable linearly ordered set endowed with its order topology. Then there is a continuous strictly increasing map $h : Y \to \mathbb{R}$.*

Proof. If Y is a singleton, take $h \equiv 0$. Otherwise, connectedness gives the two defining properties of a linear continuum: there are no adjacent points, and every nonempty subset that is bounded above has a least upper bound. To see this, adjacent points would split Y into two disjoint open rays. Also, if a nonempty set A has upper bounds but no least upper bound, the set of points that are not upper bounds of A and the set of upper bounds of A are nonempty disjoint open sets whose union is Y.

Let D be a countable dense subset of Y, and remove an endpoint of Y from D when it belongs to D; call the remaining set D°. The linear-continuum properties and the density of D imply that D° is a countable dense linear order without endpoints. In particular, between any two of its points there is another point of D°, and every one of its points has points of D° on both sides.

There is an order isomorphism

$$r : D^\circ \longrightarrow \mathbb{Q} \cap (0, 1).$$

For completeness, enumerate the two countable dense orders. Starting from the empty finite partial order isomorphism, alternately place the next unmatched point from each enumeration into its domain or range. The finitely many points already matched determine a

nonempty open interval in the other order, and density supplies an unused point in that interval. The union of the finite partial maps is an order isomorphism r.

For $y \in Y$, define

$$h(y) = \sup\{r(d) : d \in D^\circ,\ d < y\},$$

where the supremum of the empty set is 0. For $d \in D^\circ$, the definition and the density of $\mathbb{Q} \cap (0,1)$ give $h(d) = r(d)$. If $y < z$, choose $d_0, d_1 \in D^\circ$ with $y < d_0 < d_1 < z$. Then

$$h(y) \leq r(d_0) < r(d_1) \leq h(z),$$

so h is strictly increasing.

It remains to show that $h(Y)$ has no gaps. Let $a < b$ lie in $h(Y)$, let $t \in (a,b)$, and set

$$A_t = \{y \in Y : h(y) < t\}.$$

This set is nonempty and bounded above, so let $y^* = \sup A_t$. If $h(y^*) < t$, choose $d \in D^\circ$ with $h(y^*) < r(d) < t$. Then $h(d) = r(d) < t$, so $d \in A_t$, while $h(d) > h(y^*)$ implies $d > y^*$, contradicting that y^* is an upper bound of A_t. If $h(y^*) > t$, choose $d \in D^\circ$ with $t < r(d) < h(y^*)$. Then d is an upper bound of A_t, but $d < y^*$, contradicting the leastness of y^*. Therefore $h(y^*) = t$, and $h(Y)$ is an interval of $\mathbb{R}$. Hence h is an order isomorphism from Y onto an interval of $\mathbb{R}$. An order isomorphism between linearly ordered spaces with their order topologies is a homeomorphism, so h is continuous. □

Now we can complete the proof of the proposition.

Proof. Let $x \sim y$ mean that $x \succeq y$ and $y \succeq x$, and let $Y = X/\sim$ be the set of indifference classes. Define an order on Y by

$$[x] \preceq_Y [y] \quad \Longleftrightarrow \quad y \succeq x.$$

Completeness and transitivity make this a linear order. Endow Y with its order topology and define $q : X \to Y$ by $q(x) = [x]$.

For every $a \in X$, preference continuity makes the weak upper and lower contour sets closed. Indeed, the sequential definition of continuity in the notes gives closedness because X is metric. By completeness, their complements are the strict contour sets. Thus

$$q^{-1}\big(\{c \in Y : c < [a]\}\big) = \{x \in X : a \succ x\}$$

and

$$q^{-1}\big(\{c \in Y : [a] < c\}\big) = \{x \in X : x \succ a\}$$

are open. Such open rays form a subbasis for the order topology; hence q is continuous.

The space $X = \mathbb{R}^L_+$ is connected and separable, with $\mathbb{Q}^L_+$ as a countable dense subset. Since q is continuous and onto, $Y = q(X)$ is connected. Moreover, $q(\mathbb{Q}^L_+)$ is a countable dense subset of Y: the inverse image of every nonempty open set in Y is a nonempty open set in X and therefore meets $\mathbb{Q}^L_+$. The lemma therefore supplies a continuous strictly increasing map $h : Y \to \mathbb{R}$. Define

$$u = h \circ q.$$

The function u is continuous, and, for every $x, y \in X$,

$$x \succeq y \iff [x] \geq_Y [y] \iff h([x]) \geq h([y]) \iff u(x) \geq u(y).$$

Thus u represents the preferences. □

2.2.4 Existence of a Solution to the Choice Problem

Proof. Clearly if $C(\succeq, X)$ is compact, as a subset of $\mathbb{R}$ it must have a maximal element. By definition as a subset of a bounded set X, $C(\succeq, X)$ is bounded. Suppose $C(\succeq, X)$ is not compact, then there exists a sequence $\{x^k\} \subset C(\succeq, X)$ such that $\lim x^k = x \notin C(\succeq, X)$. Note by the compactness of X, $x \in X$. Thus, since $x \in X \setminus C(\succeq, X)$ it must be that for all k, $x^k \succ x$. Now we take $y \in C(\succeq, X)$ and note that $x^k \succeq y$ for all k. By preference continuity $\lim x^k \succeq y$, but since $\lim x^k = x \in X \setminus C(\succeq, X)$ and $y \in C(\succeq, X)$, $x \not\succeq y$, a contradiction. □

Note that this also means (from an optimization perspective) that any maximization problem that involves maximizing a continuous function on a compact space has a solution.

Corollary 2.28. *Suppose that X is compact (closed and bounded), and $u : X \to \mathbb{R}$ is continuous. Then* $\arg\max\{u(x) : x \in X\} \neq \varnothing$.

Example 2.29. On the interval $X = [0, 1]$, the continuous function $u(x) = -(x - 0.4)^2$ reaches its largest value at $x = 0.4$. The corollary rules out the possibility that the best value is only a limit that is never attained in X.

2.3 A Brief Detour to Classical Demand

Classical consumer demand is the basic application of the preceding choice theory. A consumer takes prices and wealth as given and chooses the most preferred affordable consumption bundle.

Let the consumption set be $X = \mathbb{R}^L_+$ and let the price–wealth domain be

$$\mathcal{P} = \mathbb{R}^L_{++} \times \mathbb{R}_+.$$

For $(p, w) \in \mathcal{P}$, the **budget set** is

$$B(p, w) = \{x \in \mathbb{R}^L_+ : p \cdot x \leq w\}.$$

Suppose that preferences are complete, transitive, and continuous on $\mathbb{R}^L_+$, so that a continuous utility representation u exists. The consumer's utility-maximization problem is

$$\max\{u(x) : x \in B(p, w)\}.$$

The associated **Marshallian demand correspondence** and **indirect utility function** are

$$D(p, w) = \underset{x \in B(p,w)}{\arg} u(x), \qquad v(p, w) = \max_{x \in B(p,w)} u(x).$$

Example 2.30. With two goods, prices $p = (2, 1)$, and wealth $w = 10$, the bundle $(3, 4)$ is affordable because $2(3)+4 = 10$. The budget set contains every nonnegative bundle satisfying

this inequality, and demand selects the affordable bundle or bundles that the consumer likes best.

Definition 2.31 (Local nonsatiation)**.** Preferences are **locally nonsatiated** if, for every $x \in X$ and every $\varepsilon > 0$, there is a $y \in X$ such that $\|y - x\| < \varepsilon$ and $y \succ x$.

Example 2.32. At any bundle of goods, local nonsatiation means that the consumer can find an arbitrarily nearby bundle that is better. For example, a very small additional amount of a desirable good can make a nearby bundle strictly preferred.

Definition 2.33. They are **weakly monotone** if $y \gg x$ implies $y \succ x$.

Example 2.34. If $x = (1, 2)$ and $y = (2, 3)$, then y has more of both goods. Weak monotonicity says the consumer strictly prefers y to x.

Definition 2.35. Preferences are **strongly monotone** if $y \geq x$ and $y \neq x$ imply $y \succ x$.

Example 2.36. If $x = (1, 2)$ and $y = (1, 3)$, then y has the same amount of the first good and more of the second. Strong monotonicity says the consumer strictly prefers y to x.

Strong monotonicity implies weak monotonicity, which implies local nonsatiation.

Lemma 2.37 (Budget-set properties)**.** *For every $(p, w) \in \mathcal{P}$, $B(p, w)$ is nonempty, compact, and convex.*

Proof. The zero bundle belongs to $B(p, w)$, so the set is nonempty. If $x \in B(p, w)$, then $0 \leq x_l \leq w/p_l$ for every l. Thus the set is bounded. It is closed because it is the intersection

of the closed orthant and the closed half-space $\{x : p \cdot x \leq w\}$. Hence it is compact. Finally, if $x, y \in B(p, w)$ and $\lambda \in [0, 1]$, then

$$p \cdot [\lambda x + (1 - \lambda)y] = \lambda p \cdot x + (1 - \lambda)p \cdot y \leq w,$$

so the budget set is convex. □

Theorem 2.38 (Existence and shape of Marshallian demand)**.** *If u is continuous, then $D(p, w)$ is nonempty and compact for every $(p, w) \in \mathcal{P}$. If preferences are convex, then $D(p, w)$ is convex.*

Example 2.39. Let $u(x) = x_1 + x_2$, let prices be $(1, 1)$, and let wealth be 10. Every bundle on the line $x_1 + x_2 = 10$ maximizes utility. The demand set is therefore nonempty, compact, and convex.

Proof. The budget-set lemma and the Weierstrass theorem imply that the continuous function u attains a maximum on $B(p, w)$, so $D(p, w)$ is nonempty. It is closed in the compact set $B(p, w)$ because

$$D(p, w) = \{x \in B(p, w) : u(x) = v(p, w)\}.$$

It is therefore compact.

Now suppose preferences are convex, meaning that every upper contour set is convex. Let $x, y \in D(p, w)$ and $\lambda \in [0, 1]$. The budget-set lemma gives $z = \lambda x + (1 - \lambda)y \in B(p, w)$. Convexity of preferences gives $z \succeq x$. Since x is optimal on the budget set, $x \succeq z$ as well. Thus $z \sim x$, so $z \in D(p, w)$. Hence $D(p, w)$ is convex. □

Theorem 2.40 (Basic demand and indirect-utility properties)**.** *Suppose u is continuous and preferences are locally nonsatiated. Then the following statements hold.*

1. *For every $x \in D(p, w)$ satisfies*

$$p \cdot x = w.$$

2. *For every $\lambda > 0$,*

$$D(\lambda p, \lambda w) = D(p, w), \qquad v(\lambda p, \lambda w) = v(p, w).$$

3. *v is nondecreasing in wealth and nonincreasing in each price: if $w' \geq w$, then $v(p, w') \geq v(p, w)$; if $p' \geq p$, then $v(p', w) \leq v(p, w)$.*

4. *The function v is continuous on $\mathcal{P}$, and D is upper hemicontinuous on $\mathcal{P}$.*

Example 2.41. With prices $p = (2, 1)$ and wealth $w = 10$, a locally nonsatiated consumer chooses a bundle that spends all 10 dollars. If both prices and wealth are doubled, the affordable set is unchanged, so the consumer's demand does not change.

Proof. For the first claim, suppose that $x \in D(p, w)$ and $p \cdot x < w$. Choose $\delta > 0$ so that $\delta \|p\| < w - p \cdot x$. By local nonsatiation, there is a $y \in X$ with $\|y - x\| < \delta$ and $y \succ x$. Then

$$p \cdot y \leq p \cdot x + \|p\| \|y - x\| < w,$$

so $y \in B(p, w)$, contradicting the optimality of x. Therefore the budget constraint binds.

For the second claim,

$$B(\lambda p, \lambda w) = B(p, w)$$

whenever $\lambda > 0$. Both conclusions follow immediately from the definitions of D and v. For the third claim, $B(p, w) \subseteq B(p, w')$ when $w' \geq w$, while $B(p', w) \subseteq B(p, w)$ when $p' \geq p$. Maximizing the same function over these nested feasible sets gives the stated inequalities.

To prove continuity, let $(p^n, w^n) \to (p, w)$. For all sufficiently large n, every component of p^n is bounded below by a positive constant. Hence every $x^n \in B(p^n, w^n)$ lies in one common bounded box. Choose $x^n \in D(p^n, w^n)$ along a subsequence attaining the limsup of $v(p^n, w^n)$. A further subsequence converges to some x. Passing to the limit in $p^n \cdot x^n \leq w^n$ gives $x \in B(p, w)$, and continuity gives

$$\limsup_{n\to\infty} v(p^n, w^n) = u(x) \leq v(p, w).$$

For the reverse inequality, first suppose $w > 0$ and take $x^* \in D(p, w)$. For each $\eta \in (0, 1)$, the bundle $(1 - \eta)x^*$ has strictly positive budget slack at (p, w) and therefore belongs to $B(p^n, w^n)$ for all sufficiently large n. Thus

$$\liminf_{n\to\infty} v(p^n, w^n) \geq u((1 - \eta)x^*).$$

Letting $\eta \downarrow 0$ yields $\liminf_n v(p^n, w^n) \geq v(p, w)$. If $w = 0$, the budget set is $\{0\}$ and the same componentwise bound implies that every $x^n \in B(p^n, w^n)$ converges to 0; continuity again gives $v(p^n, w^n) \to v(p, 0)$. Hence v is continuous.

Finally, let $(p^n, w^n) \to (p, w)$ and let $x^n \in D(p^n, w^n)$ with $x^n \to x$. Feasibility passes

to the limit, and continuity of both u and v gives $u(x) = v(p, w)$. Hence $x \in D(p, w)$, so D has a closed graph. The common bounded-box argument gives local boundedness, and the demand sets are compact. If D were not upper hemicontinuous, one could choose a convergent parameter sequence and demand selections outside a fixed open neighborhood of $D(p, w)$; local boundedness would give a convergent subsequence, and the closed-graph property would put its limit back in $D(p, w)$, a contradiction. □

2.4 Choice under Risk

Many strategic problems generate uncertainty over outcomes. To compare such objects, ordinary ordinal utility is not enough.

Let the set of primitive outcomes be

$$Z = \{z_1, \ldots, z_N\}.$$

Definition 2.42. A **simple lottery** is a probability vector

$$L = (p_1, \ldots, p_N), \qquad p_n \geq 0, \qquad \sum_{n=1}^{N} p_n = 1,$$

where p_n is the probability of outcome z_n.

Example 2.43. If the possible prizes are \$0 and \$100, then a ticket that pays \$100 with probability 0.25 and \$0 with probability 0.75 is the simple lottery $(0.75, 0.25)$ when the outcomes are listed in that order.

Let $\mathcal{L}$ denote the set of simple lotteries over Z.

Definition 2.44. A **compound lottery** first randomizes over lotteries and then realizes an outcome according to the selected lottery.

Example 2.45. First flip a fair coin. Heads selects a ticket that pays \$10 for sure, and tails selects a ticket that pays \$0 for sure. The selected ticket is then carried out, so this compound lottery reduces to a simple lottery paying each amount with probability 0.5.

A standard reduction assumption treats a compound lottery as equivalent to the simple lottery obtained by multiplying and summing the relevant probabilities.

Expected utility requires an additional preference condition.

Definition 2.46 (Independence)**.** Preferences on $\mathcal{L}$ satisfy the **independence axiom** if, for all lotteries L, L', L'' and all $\alpha \in (0,1)$,

$$L \succsim L' \quad \Longleftrightarrow \quad \alpha L + (1-\alpha)L'' \succsim \alpha L' + (1-\alpha)L''.$$

Example 2.47. Suppose a person prefers receiving \$10 for sure to receiving \$0 for sure. Independence says that after mixing each option with the same 50 percent chance of receiving \$5, the person must still prefer the mixture of \$10 and \$5 to the mixture of \$0 and \$5.

The common component L'' should not reverse the ranking between L and L'.

Theorem 2.48 (von Neumann–Morgenstern expected utility)**.** *Assume compound lotteries are reduced to their equivalent simple lotteries. If preferences over lotteries are complete, transitive, continuous, and satisfy independence, then there exists a utility index* $u : Z \to \mathbb{R}$

such that preferences are represented by

$$U(L) = \sum_{n=1}^{N} p_n u(z_n).$$

Moreover, any other expected-utility representation has the form

$$v(z) = au(z) + b, \qquad a > 0.$$

Example 2.49. Let the two outcomes be a \$100 prize and no prize, with $u(\text{prize}) = 1$ and $u(\text{no prize}) = 0$. A lottery that pays the prize with probability 0.3 has expected utility $0.3(1) + 0.7(0) = 0.3$. Under the theorem's assumptions, the decision maker ranks lotteries by calculations of this form.

The positive-affine uniqueness is stronger than ordinary ordinal uniqueness. A general increasing transformation would change expected-utility comparisons because expectation does not commute with arbitrary nonlinear transformations.

Proof. Because Z is finite and preferences are complete and transitive, choose a best outcome z^+ and a worst outcome z^-, so that

$$z^+ \succeq z \succeq z^- \qquad \text{for every } z \in Z.$$

For $\alpha \in [0, 1]$, write

$$M(\alpha) = \alpha z^+ + (1 - \alpha) z^-.$$

First suppose $z^+ \succ z^-$. Independence implies that $M(\alpha) \succ M(\beta)$ whenever $\alpha > \beta$. To see this, write $M(\alpha)$ as a nontrivial mixture of z^+ and $M(\beta)$, and use independence to preserve the strict ranking $z^+ \succ M(\beta)$. The standard substitution implication of independence will also be used below: if $L_i \sim L_i'$ for finitely many lotteries, then replacing each L_i by L_i' inside a compound lottery leaves the resulting lottery indifferent. This follows by applying independence one replacement at a time and then reducing the compound lottery.

Fix $z \in Z$ and define

$$A_z^+ = \{\alpha \in [0,1] : M(\alpha) \succeq z\}, \qquad A_z^- = \{\alpha \in [0,1] : z \succeq M(\alpha)\}.$$

The first set contains 1, the second contains 0, and both are closed by continuity. Completeness implies that they cover $[0,1]$. Since $[0,1]$ is connected, they intersect. Therefore there is a scalar $u(z) \in [0,1]$ such that

$$z \sim M(u(z)).$$

The strict ordering of the $M(\alpha)$ lotteries makes this scalar unique.

Now let $L = (p_1, \ldots, p_N)$. Repeated substitution and reduction of compound lotteries give

$$\begin{aligned} L &\sim \sum_{n=1}^{N} p_n \left[u(z_n) z^+ + (1 - u(z_n)) z^-\right] \\ &\sim \left(\sum_{n=1}^{N} p_n u(z_n)\right) z^+ + \left(1 - \sum_{n=1}^{N} p_n u(z_n)\right) z^-. \end{aligned}$$

Thus two lotteries are ranked exactly as their coefficients on z^+ in this final mixture. For

$L' = (p'_1, \ldots, p'_N)$, consequently,

$$L \succeq L' \quad \Longleftrightarrow \quad \sum_{n=1}^{N} p_n u(z_n) \geq \sum_{n=1}^{N} p'_n u(z_n),$$

which proves the expected-utility representation.

For uniqueness, let v be another expected-utility index. Since $z \sim M(u(z))$, the expectation formula for v gives

$$v(z) = u(z)v(z^+) + [1 - u(z)]v(z^-) = au(z) + b,$$

where $a = v(z^+) - v(z^-) > 0$ and $b = v(z^-)$. If instead $z^+ \sim z^-$, every outcome and every lottery is indifferent; then the constant index $u \equiv 0$ represents preferences, and every other expected-utility index is constant and hence has the same positive-affine form. □

2.5 Risk Attitudes

Suppose outcomes are monetary wealth and u is differentiable.

A decision maker is risk averse when

$$u(\mathbb{E}[w]) \geq \mathbb{E}[u(w)]$$

for every relevant wealth lottery. Concavity of u is the standard representation of risk aversion.

Example 2.50. With $u(w) = \sqrt{w}$, compare a certain \$100 to a lottery that pays \$50 or

\$150 with equal probability. Both have expected wealth \$100, but the certain amount has utility 10, while the lottery has expected utility $(\sqrt{50} + \sqrt{150})/2$, which is lower.

If $u''(w) < 0$, marginal utility declines with wealth. A certain amount equal to expected wealth can then be preferred to a risky lottery with the same mean.

Two useful local measures are absolute and relative risk aversion:

$$A(w) = -\frac{u''(w)}{u'(w)}, \qquad R(w) = -w\frac{u''(w)}{u'(w)}.$$

These quantities are invariant to positive-affine transformations and therefore are meaningful properties of a von Neumann–Morgenstern representation.

Example 2.51. For $u(w) = \log w$, absolute risk aversion is $A(w) = 1/w$, so it falls as wealth rises, while relative risk aversion is $R(w) = 1$ at every positive wealth level.

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2.6 Expanded Worked Examples

Example 2.52 (Ordinal and cardinal representations)**.** If a consumer ranks pasta above tacos above salad, both $u = (3, 2, 1)$ and $v = (27, 8, 1)$ represent the same ordinal ranking. The numerical gaps have no independent meaning. By contrast, for lotteries over \$0, \$50, and \$100, let $u = (0, 1, 2)$. A sure \$50 and an equal-probability lottery over \$0 and \$100 both have expected utility 1. The positive-affine change $v = 3u + 7$ preserves this equality, while the nonlinear change u^2 does not. This is why expected utility is cardinal only up to positive-affine changes.

Example 2.53 (Completeness, transitivity, and finite choice)**.** Suppose a student ranks apartments as $A \succ B \sim C \succ D$. Completeness means that every pair can be compared, including B and C; transitivity rules out a cycle such as $A \succ B$, $B \succ C$, and $C \succ A$. On this finite set, the choice correspondence is $\{A\}$. If instead $B \sim C \succ A \succ D$, the choice correspondence is $\{B, C\}$: utility maximization may produce more than one optimal alternative.

Example 2.54 (Continuity)**.** Let $u(x) = x_1 + x_2$ on $[0, 1]^2$. If $x^n = (1, 1/n)$ and $y^n = (0, 1)$, then $x^n \succeq y^n$ for every n, and the limiting bundles $(1, 0)$ and $(0, 1)$ are indifferent. In contrast, lexicographic preferences violate continuity: $(1/n, 0)$ is preferred to $(0, 100)$ for every n, but the limit $(0, 0)$ is not preferred to $(0, 100)$. Continuity prevents such preference jumps.

Example 2.55 (Compactness and utility maximization)**.** Let $X = \{x \in \mathbb{R}^2_+ : x_1 + x_2 \leq 10\}$ and $u(x) = \sqrt{x_1 x_2}$. The feasible set is compact and the utility function is continuous. By the arithmetic–geometric mean inequality, utility is maximized at $(5, 5)$. Compactness matters:

on $[0, 1)$, the continuous function $u(x) = x$ has no maximizer, even though its supremum is 1.

Example 2.56 (Cobb–Douglas demand)**.** For $u(x_1, x_2) = \sqrt{x_1 x_2}$, prices $(1, 1)$, and wealth 10, the consumer chooses $(5, 5)$. The budget binds. If both prices and wealth double, demand remains $(5, 5)$. If the price of good 1 rises to 2 while wealth stays 10, demand becomes $(2.5, 5)$. This illustrates budget exhaustion, degree-zero homogeneity, and the adverse effect of a price increase on the consumer's feasible opportunities.

Example 2.57 (Perfect complements and monotonicity)**.** For $u(x_1, x_2) = \min\{x_1, x_2\}$, moving from $(2, 2)$ to $(3, 3)$ raises utility, so the preference is weakly monotone. But $(2, 5)$ is indifferent to $(2, 2)$, even though it has more of good 2. Thus the preference is not strongly monotone. Its indifference curves are L-shaped, making the unused extra units visually clear.

Example 2.58 (Lotteries and risk aversion)**.** Consider a lottery that yields \$50 or \$150 with equal probability. For $u(w) = \sqrt{w}$, its expected utility is $\frac{1}{2}\sqrt{50} + \frac{1}{2}\sqrt{150} \approx 9.66$, below the utility 10 of receiving \$100 for certain. The certainty equivalent is about \$93.30, so the risk premium is about \$6.70. The chord joining the utility levels at 50 and 150 lies below the concave utility curve, which is the graphical source of risk aversion.

Chapter 3

Foundations of Game Theory: Normal Form

Individual choice theory asks what one decision maker should do when the feasible set and consequences are taken as given. Game theory begins when consequences depend on the choices of several decision makers.

A firm's profit depends on competitors. A bidder's payoff depends on other bids. A country's trade policy matters partly because other countries respond. In each case, an individual's best choice cannot be evaluated without forming a view about the behavior of others.

The normal form compresses a strategic problem into three objects: players, strategies, and payoffs. This compression is powerful, but it is important to remember what has been compressed. Timing, information, physical actions, and consequences are represented indirectly through the strategy sets and payoff functions.

3.1 The Normal Form

Definition 3.1 (Normal Form)**.** A **normal-form game** is

$$G = (N, \{S_i\}_{i \in N}, \{u_i\}_{i \in N}),$$

where $N = \{1, \ldots, n\}$ is the set of players, S_i is Player i's strategy set, and

$$u_i : S \to \mathbb{R}, \qquad S = \prod_{j=1}^{n} S_j,$$

is Player i's payoff representation.

Example 3.2 (Normal-form representation)**.** Suppose Players 1 and 2 each choose either L or R simultaneously. Then $N = \{1, 2\}$, $S_1 = S_2 = \{L, R\}$, and the four strategy profiles are (L, L), (L, R), (R, L), and (R, R). A normal form records a payoff pair for each of these four profiles. It does not need to display the physical details of the interaction; those details are summarized by the strategies and payoff functions.

A strategy profile is

$$s = (s_1, \ldots, s_n).$$

We write s_{-i} for the strategies chosen by everyone other than Player i.

A strategy is not necessarily a single physical action. It is a complete contingent plan.

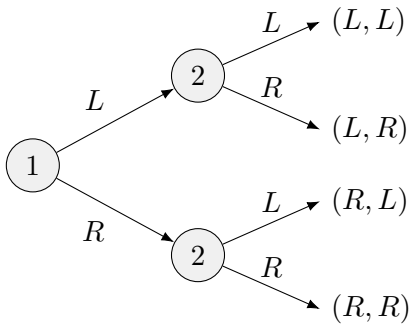
Example 3.3 (Timing changes strategies)**.** Suppose two players can choose L or R. If they move simultaneously, each has strategy set $S_i = \{L, R\}$. The normal form contains the four action profiles (L, L), (L, R), (R, L), and (R, R).

Now change only the timing. Player 1 chooses first, Player 2 observes that choice, and then Player 2 chooses. Player 1 still has $S_1 = \{L, R\}$, but Player 2 must specify a complete contingent plan. Thus

$$S_2 = \{(L, L), (L, R), (R, L), (R, R)\},$$

where (L, R) means choose L after observing L and choose R after observing R. For instance, if Player 1 chooses L and Player 2 uses (R, L), the realized outcome is (L, R). The second component of Player 2's plan is off the realized path, but it is still part of the strategy.

Player 2 observes Player 1's action



The underlying physical actions can therefore remain unchanged while the strategy space changes because timing or information changes.

3.2 Outcomes, Preferences, and Payoffs

A payoff function is attached to strategy profiles, but conceptually it represents preferences over the consequences induced by those profiles. Two environments with identical physical outcomes can be different games if the players evaluate those outcomes differently.

Example 3.4 (The same prison terms, different preferences)**.** Suppose the physical prison

sentences generated by C and D are

	C	D
C	$(1, 1)$	$(3, 0)$
D	$(0, 3)$	$(2, 2)$

where smaller sentences are physically better for the prisoner receiving them.

If each player cares only about her own sentence, a payoff representation is

	C	D
C	$(-1, -1)$	$(-3, 0)$
D	$(0, -3)$	$(-2, -2)$.

This is a Prisoner's Dilemma.

Now suppose Player 1 cares about the sum of both prison sentences while Player 2 still cares only about her own. Player 1's payoffs become $-2, -3, -3, -4$ across the four outcomes. The physical sentencing technology has not changed, but the strategic game has.

Game theory therefore does not assume that preferences are selfish. Altruism, spite, fairness concerns, social preferences, or concern about the welfare of others can all be represented if they are part of the player's preferences.

3.3 Ordinal and Cardinal Payoffs

For deterministic pure-strategy comparisons, any strictly increasing transformation

$$v_i = f_i \circ u_i$$

preserves best responses and pure-strategy Nash equilibria. Pure-strategy analysis is therefore ordinal.

Once a player evaluates probabilistic beliefs or mixed strategies, however, the player is comparing lotteries. Expected payoff calculations use the von Neumann–Morgenstern structure from Chapter 2. The permissible transformation is then

$$v_i = a_i u_i + b_i, \qquad a_i > 0.$$

This distinction is easy to overlook and becomes important whenever expected payoffs are used.

3.4 Dominance

Definition 3.5 (Strictly dominated strategy)**.** A strategy s_i is **strictly dominated** by s_i' if

$$u_i(s_i', s_{-i}) > u_i(s_i, s_{-i})$$

for every $s_{-i} \in S_{-i}$.

Example 3.6. A firm chooses either *Stay* in a market or *Exit*, while its rival chooses a high or low price. Suppose the firm's own profits are

	Rival: High	Rival: Low
Stay	5	2
Exit	0	0.

Staying is better in both columns: $5 > 0$ when the rival charges a high price and $2 > 0$ when it charges a low price. Exit is therefore strictly dominated by Stay. The comparison does not use a probability assessment about the rival; it holds for every possible rival action.

A strictly dominated strategy is worse regardless of what the other players do. If a player is rational and understands her own payoff function, such a strategy should not be chosen.

Definition 3.7 (Strictly dominant strategy)**.** A strategy s_i^* is **strictly dominant** if it strictly dominates every other strategy in S_i.

Example 3.8. In the Prisoner's Dilemma below, suppose Player 2 cooperates. Player 1 then receives 5 from choosing D and 3 from choosing C. If Player 2 defects, Player 1 receives 1 from D and 0 from C. Thus D is better in every contingency. The same comparison applies to Player 2, so D is a strictly dominant strategy for each player; no belief about the other player's choice is required.

Example 3.9 (Prisoner's Dilemma)**.** Consider

	C	D
C	$(3,3)$	$(0,5)$
D	$(5,0)$	$(1,1)$.

For each player, D strictly dominates C. The outcome (D, D) therefore follows from individual dominance reasoning. Yet (C, C) gives both players a larger payoff. Individual rationality does not by itself imply social efficiency.

This distinction anticipates the market chapters. Equilibrium is a positive solution concept. Efficiency is a separate normative property.

3.5 Iterated Deletion of Strictly Dominated Strategies

Dominance reasoning can be applied repeatedly.

Definition 3.10. Iterated deletion of strictly dominated strategies begins with $S_i^0 = S_i$ and repeatedly removes strategies that are strictly dominated relative to the strategies still available to the other players.

Example 3.11. In the matrix below, Player 1's payoff from U exceeds the payoff from D in every column: $3 > 2$, $2 > 1$, and $1 > 0$. First delete D. Player 2 then knows that only the U row remains relevant. Along that row, Player 2 receives 2 from L, 1 from M, and 0 from R, so both M and R can next be deleted. The surviving outcome (U, L) is obtained by successive rounds of this reasoning, rather than by a single comparison in the original game.

Example 3.12. Consider

	L	M	R
U	$(3,2)$	$(2,1)$	$(1,0)$
D	$(2,2)$	$(1,3)$	$(0,4)$.

For Player 1, U strictly dominates D. After deleting D, Player 2 compares payoffs only against U and prefers L to both M and R. The unique surviving profile is (U, L).

The second deletion uses a higher-order inference: Player 2 reasons that Player 1 will not use a strictly dominated strategy. Repeated deletion therefore embodies increasingly strong assumptions about mutual rationality.

For strict dominance in finite games, deletion order does not create the difficulties that can arise under weak dominance.

3.6 Beliefs and Rationalizability

Dominance is demanding because a strategy must be inferior against every opponent behavior. A weaker idea asks whether a strategy can be optimal under some reasonable belief about opponents.

Let

$$\mu_i \in \Delta(S_{-i})$$

be a probability distribution over opponents' strategy profiles. Once beliefs are probabilistic, expected utility is being used.

Definition 3.13 (Best response to a belief)**.** A strategy s_i is a best response to belief μ_i if

$$s_i \in \arg\max_{s_i' \in S_i} \sum_{s_{-i}} \mu_i(s_{-i}) u_i(s_i', s_{-i}).$$

Example 3.14. Suppose a firm believes that its rival chooses a high price with probability $q = 3/4$. Its profit from choosing High is 10 when the rival chooses High and 2 when the rival chooses Low; its profit from choosing Low is 5 when the rival chooses High and 7 when the rival chooses Low. Hence

$$\mathbb{E}[\pi(\text{High})] = \tfrac{3}{4}(10) + \tfrac{1}{4}(2) = 8, \qquad \mathbb{E}[\pi(\text{Low})] = \tfrac{3}{4}(5) + \tfrac{1}{4}(7) = 5.5.$$

High is a best response to this belief. The calculation identifies a best response to $q = 3/4$; it does not assert that the rival will actually choose High with that probability.

Definition 3.15 (Rationalizable strategy)**.** Let $R_i^0 = S_i$. Recursively, R_i^{k+1} contains the strategies in R_i^k that are best responses to some belief supported on R_{-i}^k. The set of **rationalizable strategies** is

$$R_i^\infty = \bigcap_{k=0}^{\infty} R_i^k.$$

Example 3.16. Consider an entrant choosing *Enter* or *Out* and an incumbent choosing

Accommodate or *Fight*. Let the payoff matrix be

	Accommodate	Fight
Enter	$(3, 2)$	$(-1, -1)$
Out	$(0, 4)$	$(0, 4)$.

If the entrant assigns probability r to accommodation, Enter yields $3r-(1-r) = 4r-1$, while Out yields 0. Thus Enter is a best response for every belief with $r \geq 1/4$. Because neither incumbent action is removed by strict-dominance reasoning, such a belief is permitted and Enter is rationalizable. The belief need not be the one that is actually correct.

For games with more than two players, a belief over the joint strategy profile of the opponents can allow correlation among their behavior.

The epistemic idea is mutual rationality: I am rational; I believe the others are rational; I believe that they believe I am rational; and so on. Rationalizability does not require the player's beliefs to be correct. It requires only that the strategy be optimal under some belief consistent with the successive restrictions.

3.7 Nash Equilibrium

Definition 3.17 (Best response)**.** Player i's best-response correspondence is

$$\mathrm{BR}_i(s_{-i}) = \arg\max_{s_i \in S_i} u_i(s_i, s_{-i}).$$

Example 3.18. Suppose two coffee shops choose High or Low prices, with payoffs

	High	Low
High	$(5,5)$	$(1,6)$
Low	$(6,1)$	$(3,3)$.

Against the rival's High price, a shop gets 5 from High and 6 from Low, so its best response is Low. Against the rival's Low price, it gets 1 from High and 3 from Low, so Low is again the best response. This table makes the correspondence explicit: $BR_i(\text{High}) = \{\text{Low}\}$ and $BR_i(\text{Low}) = \{\text{Low}\}$.

Definition 3.19 (Pure-strategy Nash equilibrium)**.** A strategy profile s^* is a **Nash equilibrium** if

$$s_i^* \in \text{BR}_i(s_{-i}^*)$$

for every player i. Equivalently,

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*)$$

for every $s_i \in S_i$ and every i.

Example 3.20. Return to the Prisoner's Dilemma. At (D, D), a player who switches alone from D to C lowers her payoff from 1 to 0. Hence (D, D) is a Nash equilibrium: neither player has a profitable unilateral deviation. Nevertheless, both players receive 3 at (C, C), which is larger than 1. The move from (D, D) to (C, C) requires coordinated changes by both players, so it does not contradict Nash equilibrium.

A Nash equilibrium rules out profitable *unilateral* deviations. It does not require the outcome to maximize joint payoffs, and it does not rule out profitable coordinated deviations by coalitions.

Nash equilibrium also imposes a consistency requirement absent from rationalizability. Each player's strategy must be optimal against what the other players actually do in equilibrium.

Theorem 3.21 (Existence of pure-strategy Nash equilibrium)**.** *Let* $G = (N, \{S_i\}_{i\in N}, \{u_i\}_{i\in N})$ *be a finite-player normal-form game. Suppose each* S_i *is a nonempty, compact, convex subset of a finite-dimensional Euclidean space, each* u_i *is continuous, and* $u_i(\cdot, s_{-i})$ *is quasi-concave on* S_i *for every* s_{-i}*. Then* G *has a pure-strategy Nash equilibrium.*

Proof. For each player, define

$$BR_i(s_{-i}) = \arg\max_{s_i \in S_i} u_i(s_i, s_{-i}).$$

Compactness and continuity imply that BR_i is nonempty and compact valued; quasi-concavity implies convex values; and the maximum theorem implies upper hemicontinuity. Hence $BR(s) = \prod_i BR_i(s_{-i})$ is an upper hemicontinuous correspondence from the nonempty compact convex set $S = \prod_i S_i$ into itself with nonempty compact convex values. Kakutani's fixed-point theorem gives $s^* \in BR(s^*)$, which is a Nash equilibrium. □

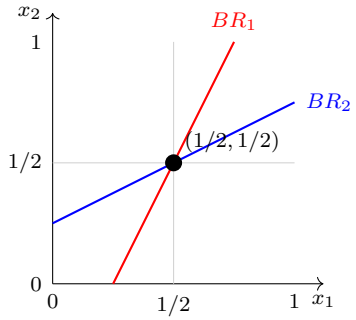
Example 3.22 (Best-response curves and the fixed-point intuition)**.** Let $x_1, x_2 \in [0, 1]$ and let

$$u_1(x_1, x_2) = -\left[x_1 - \left(\tfrac{1}{4} + \tfrac{1}{2}x_2\right)\right]^2, \qquad u_2(x_1, x_2) = -\left[x_2 - \left(\tfrac{1}{4} + \tfrac{1}{2}x_1\right)\right]^2.$$

Each payoff is continuous and strictly concave in the player's own action. The best responses are $BR_1(x_2) = \{\frac{1}{4} + \frac{1}{2}x_2\}$ and $BR_2(x_1) = \{\frac{1}{4} + \frac{1}{2}x_1\}$. Their intersection solves

$$x_1 = \tfrac{1}{4} + \tfrac{1}{2}x_2, \qquad x_2 = \tfrac{1}{4} + \tfrac{1}{2}x_1,$$

so the equilibrium is $(x_1^*, x_2^*) = (\frac{1}{2}, \frac{1}{2})$.



In this single-valued example a fixed point is literally an intersection of two curves. In general, best responses may be sets, and Kakutani supplies the analogous fixed point for the set-valued correspondence.

Proposition 3.23. *A strictly dominated strategy cannot be used as a component of a Nash equilibrium.*

Example 3.24. In the Prisoner's Dilemma, C is strictly dominated by D. If a proposed profile has Player 1 choose C, then switching to D raises Player 1's payoff whether Player 2 chooses C ($5 > 3$) or D ($1 > 0$). Thus neither (C, C) nor (C, D) can be a Nash equilibrium; analogous reasoning eliminates profiles with Player 2 choosing C. The only possible equilibrium is (D, D).

Proof. If s_i is strictly dominated by s_i', then

$$u_i(s_i', s_{-i}) > u_i(s_i, s_{-i})$$

for every s_{-i}. In particular the inequality holds at any proposed equilibrium opponents' profile, so s_i cannot be a best response. □

Proposition 3.25. *For every player i, every strategy that appears as a component of a pure-strategy Nash equilibrium is rationalizable:*

$$\mathrm{proj}_i\, NE(G) \subseteq R_i^\infty.$$

Example 3.26. In the coordination game below, U is used in the equilibrium (U, L). It is a best response to the degenerate belief assigning probability one to Player 2's choice of L, because $2 > 0$. Similarly, L is a best response to the belief assigning probability one to U. The other equilibrium (D, R) provides corresponding beliefs supporting D and R. Every action therefore survives rationalizability, which illustrates why rationalizability need not select a unique equilibrium.

The reason is that an equilibrium strategy is a best response to the degenerate belief assigning probability one to the opponents' equilibrium strategies. Those strategies survive the same reasoning at every round.

Example 3.27 (Coordination)**.** Consider

	L	R
U	$(2,2)$	$(0,0)$
D	$(0,0)$	$(1,1)$.

Both (U, L) and (D, R) are Nash equilibria. All four individual actions are rationalizable. Rationalizability alone therefore does not select a profile, and Nash equilibrium can itself be multiple.

Multiplicity is not merely a theoretical nuisance. It becomes a major empirical problem when different equilibria imply different observable outcomes.

3.8 Pure Equilibrium Need Not Exist

Example 3.28 (No pure equilibrium in Matching Pennies)**.** Consider Matching Pennies:

	H	T
H	$(1,-1)$	$(-1,1)$
T	$(-1,1)$	$(1,-1)$.

At every pure strategy profile, one player wants to change actions. There is no pure-strategy Nash equilibrium. For example, at (H, H) Player 2 changes to T to raise her payoff from -1 to 1; at (H, T) Player 1 changes to T to raise his payoff from -1 to 1. The analogous deviations occur at the remaining two profiles.

This motivates mixed strategies.

3.9 Mixed Strategies

Definition 3.29 (Mixed strategy)**.** A **mixed strategy** for Player i is a probability distribution

$$\sigma_i \in \Delta(S_i).$$

A pure strategy is the special case assigning probability one to a single element of S_i.

Example 3.30. A player who chooses H with probability 0.6 and T with probability 0.4 uses the mixed strategy $\sigma_i = (0.6, 0.4)$. On a single play the realized action is either H or T; the randomization occurs before the action is realized. If this player faces an opponent who chooses Heads with probability q, then the player's expected payoff from a particular mixed strategy is the probability-weighted average of the payoffs generated by its two pure actions.

If players randomize independently, expected payoff is

$$U_i(\sigma) = \sum_{s \in S} \left[\prod_{j=1}^{n} \sigma_j(s_j) \right] u_i(s).$$

Expected payoff is linear in each player's own mixed strategy holding the others fixed.

Definition 3.31 (Mixed-strategy Nash equilibrium)**.** A mixed profile σ^* is a Nash equilibrium if

$$U_i(\sigma_i^*, \sigma_{-i}^*) \geq U_i(\sigma_i, \sigma_{-i}^*)$$

for every $\sigma_i \in \Delta(S_i)$ and every player i.

Theorem 3.32 (Glicksberg: existence of mixed-strategy Nash equilibrium)**.** *Let G be a finite-player normal-form game in which every pure-strategy set S_i is a nonempty compact metric space and every payoff function u_i is continuous. Then G has a mixed-strategy Nash equilibrium.*

Proof intuition. Passing from S_i to the set $\Delta(S_i)$ of probability distributions convexifies the strategy space. Under the weak-convergence topology, $\Delta(S_i)$ is compact and convex. Expected payoff is continuous and affine in a player's own mixed strategy, so mixed best responses are nonempty, compact, convex valued, and upper hemicontinuous. Kakutani's fixed-point theorem then produces a profile at which every mixed strategy is a best response to the others. □

Example 3.33 (Matching Pennies and Glicksberg's fixed point)**.** Let p be Player 1's probability of Heads and q Player 2's probability of Heads. Player 1's expected payoffs from pure Heads and Tails are

$$U_1(H, q) = 2q - 1, \qquad U_1(T, q) = 1 - 2q.$$

Thus Player 1 chooses Heads when $q > 1/2$, Tails when $q < 1/2$, and is indifferent at $q = 1/2$. Symmetrically, Player 2 chooses Tails when $p > 1/2$, Heads when $p < 1/2$, and is indifferent at $p = 1/2$. The pure best responses do not intersect, but the mixed best-response correspondence has the fixed point $(p, q) = (1/2, 1/2)$. Each player's randomization makes the other indifferent between the two pure actions.

Proposition 3.34 (Support property)**.** *If a mixed strategy is a best response, every pure strategy used with positive probability must yield the same maximal expected payoff against*

the opponents' strategies.

Example 3.35. Suppose three pure actions have expected payoffs 6, 4, and 6 against the opponents' mixed strategies. A best response can put positive probability on the first and third actions, since each attains the maximum payoff 6. It must put probability zero on the second action: moving any positive probability from that action to either maximal action increases expected payoff. Thus a mixed best response may randomize, but only among pure best responses.

Otherwise probability could be moved away from an inferior pure strategy toward a superior one.

3.10 Matching Pennies

Let Player 1 choose H with probability p and Player 2 choose H with probability q.

Player 1's payoff from H is

$$U_1(H, q) = 2q - 1,$$

while the payoff from T is

$$U_1(T, q) = 1 - 2q.$$

For Player 1 to mix,

$$2q - 1 = 1 - 2q,$$

so

$$q = \frac{1}{2}.$$

Similarly, Player 2 is willing to mix only when

$$p = \frac{1}{2}.$$

Therefore the mixed equilibrium is

$$\boxed{p^* = q^* = \frac{1}{2}.}$$

Notice the logic: Player 1's mixing probability makes Player 2 indifferent, and Player 2's mixing probability makes Player 1 indifferent.

Mixed strategies can be interpreted as literal randomization, as population frequencies in suitable environments, or as a representation of beliefs about unpredictable behavior. Which interpretation is useful depends on the application.

3.11 The Hierarchy of Solution Concepts

The main solution concepts can be organized by the assumptions they impose.

Strict dominance says a strategy is irrational because another strategy is better regardless of beliefs. Iterated dominance adds higher-order reasoning about the rationality of others. Rationalizability permits any behavior that is optimal under some admissible belief consistent with mutual rationality. Nash equilibrium adds correct strategic conjectures: each player's behavior is optimal against the behavior the other players actually use.

None of these concepts implies efficiency. That distinction will become central when

games are used to model markets.

The existence results above show that compactness, continuity, and quasi-concavity give a pure equilibrium, while Glicksberg's theorem gives a mixed equilibrium under compactness and continuity. The next chapter asks what changes when the same players expect to interact again in the future.

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